PROJECT DESIGN PHASE-II

Data Flow Diagram & User Stories

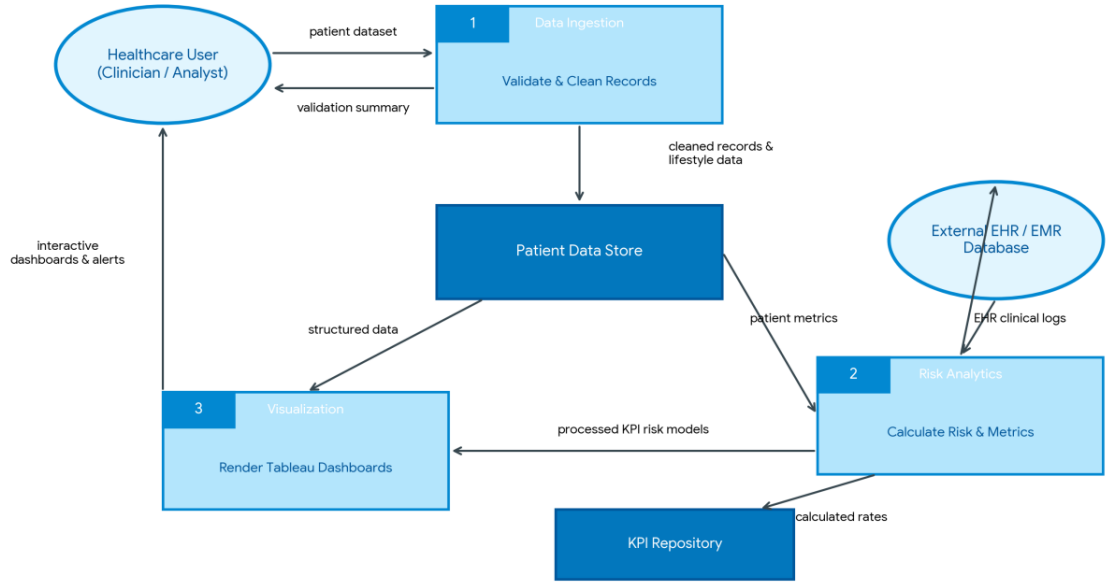
Date: 28 October 2026	**Team ID:** Individual Project
Project Name: Health360_HealthCare_Analysis	**Maximum Marks:** 4 Marks

Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Healthcare Professional, EHR System).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Data Ingestion, Risk Assessment, Visual Rendering).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Patient Healthcare Dataset, KPI Risk Repository).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example: DFD Level 0 (Industry Standard)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Clinical Risk Officer (Dr. Michael Carter)	Clinical Diagnostics	**USN-1**	As a Clinical Risk Officer, I can view overall KPI rates for Diabetes, Heart Disease, and Stroke on a single dashboard.	Dashboard displays Total Patients (5,000), Diabetes Rate (34.92%), Heart Disease Rate (29.98%), and Stroke Rate (24.86%) accurately.	High	Sprint-1
Clinical Risk Officer (Dr. Michael Carter)	Clinical Diagnostics	**USN-2**	As a Clinical Risk Officer, I can analyze Blood Pressure vs. Glucose scatter plots categorized by Heart Disease status.	Scatter plot renders 5,000 patient points with clear color separation for Heart Disease positive/negative cases.	High	Sprint-1
Clinical Risk Officer (Dr. Michael Carter)	Risk Stratification	**USN-3**	As a Clinical Risk Officer, I can filter patient cohorts by specific clinical indicators (Glucose, BP, Cholesterol).	Dynamic filters update chart visualizations in under 2 seconds.	High	Sprint-1
Clinical Risk Officer (Dr. Michael Carter)	Early Warning System	**USN-4**	As a Clinical Risk Officer, I can identify high-risk individual outliers for immediate medical	High-risk patient records are highlighted clearly on interactive scatter plots.	Medium	Sprint-2

			intervention.			
Healthcare Data Analyst (Emily Rodriguez)	Behavioral Analytics	**USN-5**	As a Healthcare Data Analyst, I can analyze disease occurrence against smoking habits across patient groups.	Bar charts compare disease prevalence between 2,538 smokers and non-smokers.	High	Sprint-1
Healthcare Data Analyst (Emily Rodriguez)	Behavioral Analytics	**USN-6**	As a Healthcare Data Analyst, I can visualize patient distribution across Exercise Levels (Level 0, Level 1, Level 2).	Pie/Donut charts display 33.5% Level 1, 33.5% Level 2, and 32.9% Level 0 distribution accurately.	High	Sprint-1
Healthcare Data Analyst (Emily Rodriguez)	Multi-Variable Correlation	**USN-7**	As a Healthcare Data Analyst, I can cross-examine alcohol consumption and fatigue patterns with chronic conditions.	Heatmaps display multi-variable interaction correlations clearly.	Medium	Sprint-2
Healthcare Data Analyst (Emily Rodriguez)	Interactive Dashboarding	**USN-8**	As a Healthcare Data Analyst, I can build interactive trend charts and comparison dashboards in Tableau.	Dashboard supports dynamic multi-parameter filtering and tooltips.	High	Sprint-1
Public Health Researcher (Dr. Priya Nair)	Demographic Profiling	**USN-9**	As a Public Health Researcher, I can view disease prevalence across different Age Groups (Teenager, Adult, Middle Age, Senior, Old).	Color intensity matrix displays age group disease distributions accurately.	High	Sprint-1
Public Health Researcher (Dr. Priya Nair)	Demographic Profiling	**USN-10**	As a Public Health Researcher, I can analyze chronic disease rates across BMI Categories (Obese, Overweight, Normal, Underweight).	Heart disease rate by BMI shows Obese (39.98%) and Overweight (32.91%) segments prominently.	High	Sprint-1
Public Health Researcher (Dr. Priya Nair)	Policy Support	**USN-11**	As a Public Health Researcher, I can identify vulnerable population segments to guide preventive health awareness programs.	Vulnerable segments (Seniors, Obese smokers) are highlighted in summary reports.	Medium	Sprint-2
System Administrator	Data Ingestion	**USN-12**	As an Administrator, I can ingest and validate 5,000 patient records from EHR systems into the data store.	Data store validates missing values and formats clinical fields correctly.	High	Sprint-1
System Administrator	Security & Access	**USN-13**	As an Administrator, I can manage user role access to ensure healthcare data privacy.	Users log in with role-based permissions (Clinician, Analyst, Administrator).	High	Sprint-1
System Administrator	Reporting & Export	**USN-14**	As an Administrator, I can export Tableau visual stories and summary reports for executive decision-making.	Reports export smoothly to PDF and interactive web views.	Low	Sprint-2